

Semantic Notions in First-Order Logic

Satisfaction · Definability · Classes of Structures

Schola Arcana · Proof Vault · Track 1

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Motivation

First-order logic has two sides: syntax and semantics. The syntactic side asks what can be *proved*. The semantic side asks what is *true*. This entry develops the semantic side: the Tarski satisfaction definition, the central semantic relations between formulas and structures, the notion of definability inside a structure, and the classification of collections of structures by their first-order expressibility. These notions are the foundation on which the completeness theorem — that semantic truth and syntactic provability coincide — will be built.

1 The Satisfaction Definition

Structures and Assignments

Definition 1 (First-order structure). A *structure* $\mathfrak{A}$ for a first-order language $\mathcal{L}$ consists of a nonempty *universe* $|\mathfrak{A}|$, together with interpretations of all nonlogical symbols: for each constant symbol c , an element $c^{\mathfrak{A}} \in |\mathfrak{A}|$; for each n -place function symbol f , a function $f^{\mathfrak{A}} : |\mathfrak{A}|^n \rightarrow |\mathfrak{A}|$; for each n -place predicate symbol P , a relation $P^{\mathfrak{A}} \subseteq |\mathfrak{A}|^n$.

Definition 2 (Variable assignment and modification). A *variable assignment* is a function $s : V \rightarrow |\mathfrak{A}|$. The *modified assignment* $s(x \mid d)$ agrees with s everywhere except at x , where it takes value $d \in |\mathfrak{A}|$:

$$s(x \mid d)(y) = \begin{cases} s(y) & \text{if } y \neq x, \\ d & \text{if } y = x. \end{cases}$$

Term Evaluation

Definition 3 (Extension of s to terms). Given $s : V \rightarrow |\mathfrak{A}|$, define $\bar{s} : T \rightarrow |\mathfrak{A}|$ by recursion:

- (1) $\bar{s}(x) = s(x)$ for each variable x .
- (2) $\bar{s}(c) = c^{\mathfrak{A}}$ for each constant symbol c .
- (3) $\bar{s}(f t_1 \cdots t_n) = f^{\mathfrak{A}}(\bar{s}(t_1), \dots, \bar{s}(t_n))$ for each n -place function symbol f .

Existence and uniqueness of $\bar{s}$ follow from the Recursion Theorem (§1.4) and unique decomposability of terms (§2.3).

The Satisfaction Relation

Definition 4 (Satisfaction, $\models_{\mathfrak{A}} \varphi[s]$). We define $\models_{\mathfrak{A}} \varphi[s]$ (read: “ $\mathfrak{A}$ satisfies φ under s ”) by recursion on formula complexity:

- (I) **Atomic — equality.** $\models_{\mathfrak{A}} t_1 t_2 [s]$ iff $\bar{s}(t_1) = \bar{s}(t_2)$.
- (II) **Atomic — predicate.** For n -place P : $\models_{\mathfrak{A}} P t_1 \cdots t_n [s]$ iff $\langle \bar{s}(t_1), \dots, \bar{s}(t_n) \rangle \in P^{\mathfrak{A}}$.

(III) **Negation.** $\models_{\mathfrak{A}} \neg \varphi[s]$ iff $\not\models_{\mathfrak{A}} \varphi[s]$.

(IV) **Implication.** $\models_{\mathfrak{A}} (\varphi \rightarrow \psi)[s]$ iff $\not\models_{\mathfrak{A}} \varphi[s]$ or $\models_{\mathfrak{A}} \psi[s]$.

(V) **Universal quantifier.** $\models_{\mathfrak{A}} \forall x \varphi[s]$ iff for every $d \in |\mathfrak{A}|$, $\models_{\mathfrak{A}} \varphi[s(x \mid d)]$.

The definition is well-founded because each clause applies to strictly simpler formulas; it bottoms out at atomic formulas, handled explicitly in (I) and (II).

Remark 1 (Why the metalanguage). The relation $\models_{\mathfrak{A}}$ is a symbol of the *metalanguage* — the informal mathematical language in which we reason about first-order logic. It does not appear inside any object-language formula. This separation, due to Tarski, prevents the Liar Paradox: no sentence of the object language can assert its own (un)satisfiability, because $\models_{\mathfrak{A}}$ is simply unavailable there.

Example 1 (Concrete satisfaction). Let $\mathfrak{A} = (\mathbb{N}; \leq, S, 0)$ where $S(n) = n + 1$. Let $s(v_1) = 0$, $s(v_2) = 1$.

$\models_{\mathfrak{A}} \forall v_1 Pcv_1[s]$ iff for every $d \in \mathbb{N}$, $\langle 0^{\mathfrak{A}}, d \rangle \in P^{\mathfrak{A}}$, i.e. $0 \leq d$ for all $d \in \mathbb{N}$. True. ✓

$\not\models_{\mathfrak{A}} \forall v_1 Pv_2v_1[s]$ because taking $d = 0$ gives $\langle s(v_2), 0 \rangle = \langle 1, 0 \rangle \notin P^{\mathfrak{A}}$, i.e. $1 \leq 0$ is false. ✓

2 Semantic Relations Between Formulas

Definition 5 (Logical implication and validity). Let Γ be a set of wffs and φ a wff.

- $\Gamma \models \varphi$ (Γ *logically implies* φ) iff for every structure $\mathfrak{A}$ and every assignment s : if $\models_{\mathfrak{A}} \gamma[s]$ for all $\gamma \in \Gamma$, then $\models_{\mathfrak{A}} \varphi[s]$.
- $\models \varphi$ (φ *is valid*) iff $\emptyset \models \varphi$, i.e. $\mathfrak{A}$ satisfies φ under every s for every $\mathfrak{A}$.
- $\varphi \models \psi$ (*logical equivalence*) iff $\varphi \models \psi$ and $\psi \models \varphi$.

Corollary 1 (Sentences, Corollary 22C). *For a set Σ of sentences and a sentence τ : $\Sigma \models \tau$ iff every model of Σ is a model of τ . A sentence τ is valid iff it is true in every structure.*

Example 2 (Key logical implications). The following hold (readers are invited to verify each):

1. $\forall v_1 Qv_1 \models Qv_2$.
2. $Qv_1 \not\models \forall v_1 Qv_1$ (counterexample: domain $\{a, b\}$, $Q^{\mathfrak{A}} = \{a\}$, $s(v_1) = a$).
3. $\models \neg \neg \sigma \rightarrow \sigma$.
4. $\forall v_1 Qv_1 \models \exists v_2 Qv_2$ (every structure has a nonempty universe).
5. $\exists x \forall y Pxy \models \forall y \exists x Pxy$ but not conversely.
6. $\models \exists x (Qx \rightarrow \forall x Qx)$ (strange but valid: in every structure either $\forall x Qx$ holds — making the consequent true — or some element fails Q — making the antecedent false).

3 Definability in a Structure

Definition 6 (Definable relation). Let $\mathfrak{A}$ be a structure and φ a formula whose free variables are among $v_1, \dots, v_k$. The k -ary relation defined by φ in $\mathfrak{A}$ is:

$$\{\langle a_1, \dots, a_k \rangle \mid \models_{\mathfrak{A}} \varphi[[a_1, \dots, a_k]]\} \subseteq |\mathfrak{A}|^k.$$

A k -ary relation R on $|\mathfrak{A}|$ is *definable* in $\mathfrak{A}$ iff some such formula defines it there.

Remark 2. Definability is always relative to a specific structure. The same formula may define different relations in different structures. The definable relations are exactly those the language can *see* inside $\mathfrak{A}$; everything else is invisible to it. The symmetries (automorphisms) of $\mathfrak{A}$ impose fundamental limits: if two elements are structurally indistinguishable, no formula can separate them.

Example 3 (The real field $\mathfrak{R} = (\mathbb{R}; 0, 1, +, \cdot)$). Although $<$ is not a symbol of the language:

1. The set $[0, \infty)$ is definable by $\exists v_2 (v_1 = v_2 \cdot v_2)$, because $a \geq 0$ iff a has a square root in $\mathbb{R}$.
2. The ordering relation $\leq$ is definable by $\exists v_3 (v_2 = v_1 + v_3 \cdot v_3)$, because $a \leq b$ iff $b - a \geq 0$ iff $b - a$ is a perfect square.

Example 4 (The natural number structure $\mathfrak{N} = (\mathbb{N}; 0, S, +, \cdot)$). 1. The ordering $<$ is definable by $\exists v_3 (v_1 + Sv_3 = v_2)$.

2. Every singleton $\{n\}$ is definable: $\{2\}$ by $v_1 = SS0$, and so on. Each n is a *definable element*.
3. The set of primes is definable by:

$$\exists v_3 (S0 + Sv_3 = v_1) \wedge \forall v_2 \forall v_3 (v_1 = v_2 \cdot v_3 \rightarrow v_2 = S0 \vee v_3 = S0).$$

The first conjunct says $v_1 > 1$; the second says every factorisation is trivial.

Remark 3 (Undefinable sets exist but are hard to exhibit). There are uncountably many subsets of $\mathbb{N}$ but only countably many formulas, so most subsets are undefinable. A specific undefinable set — the set of Gödel numbers of sentences true in $\mathfrak{N}$ — appears in §3.5 as a consequence of the incompleteness theorem.

Example 5 (Directed graph — limits of definability). Let $\mathfrak{A} = (\{a, b, c\}; \{\langle a, b \rangle, \langle a, c \rangle\})$. The set $\{b, c\}$ is definable by $\exists v_2 Ev_2 v_1$ (“something points to v_1 ”). The set $\{b\}$ is *not* definable: b and c are structurally indistinguishable (same incoming and outgoing edges), so no formula can separate them. Proof uses the homomorphism theorem (§2.5).

4 Definability of Classes of Structures

Definition 7 (Mod Σ). For a set of sentences Σ , let Mod Σ denote the class of all structures in which every member of Σ is true. Σ acts as a *filter* on the universe of all structures. For a single sentence τ we write Mod τ for Mod $\{\tau\}$.

Definition 8 (EC and EC $_{\Delta}$). A class $\mathcal{K}$ of structures is:

- an *elementary class* (**EC**) iff $\mathcal{K} = \text{Mod } \tau$ for some single sentence τ .
- an *elementary class in the wider sense* (**EC $_{\Delta}$**) iff $\mathcal{K} = \text{Mod } \Sigma$ for some set of sentences Σ .

Every EC class is EC_Δ ; the converse fails.

Example 6 (EC classes). 1. **Graphs.** Defined by the single sentence:

$$\forall x(\neg xEx \wedge \forall y(xEy \rightarrow yEx)).$$

Irreflexivity and symmetry conjoined. One sentence. EC. ✓

2. **Ordered sets.** Defined by the conjunction of: transitivity $\forall x\forall y\forall z(xPy \rightarrow yPz \rightarrow xPz)$, trichotomy $\forall x\forall y(xPy \vee x=y \vee yPx)$, and asymmetry $\forall x\forall y(xPy \rightarrow \neg yPx)$. One sentence (their conjunction). EC. ✓

3. **Groups.** Defined by the conjunction of the three group axioms (associativity, right-division, left-division). EC. ✓

Example 7 (EC_Δ but not EC — infinite groups). Define sentences expressing “there are at least n elements”:

$$\lambda_n = \exists x_1 \cdots \exists x_n \left(\bigwedge_{1 \leq i < j \leq n} x_i \neq x_j \right).$$

Then $\Sigma = \{\text{group axioms}\} \cup \{\lambda_2, \lambda_3, \dots\}$ satisfies $\text{Mod } \Sigma = \text{class of infinite groups}$: a group is infinite iff it satisfies every λ_n . Any finite group of size k fails λ_{k+1} and is excluded. Hence the class of infinite groups is EC_Δ .

It is *not* EC: no single sentence can separate infinite groups from finite ones. Proof via compactness (§2.6).

Example 8 (Not even EC_Δ — finite graphs). The class of all finite graphs is not EC_Δ . No set of sentences, however large, defines exactly the finite structures.

Proof sketch (via compactness, proved in §2.6). Suppose for contradiction that Σ defines the finite graphs. For each n , $\lambda_n \in \Sigma$ is false in some finite graph, so $\Sigma \cup \{\lambda_n\}$ is consistent for each n . By compactness, $\Sigma \cup \{\lambda_2, \lambda_3, \dots\}$ has a model — an infinite graph satisfying Σ , contradicting that Σ defines only finite graphs.

Conclusion. Finiteness is not expressible in FOL.

Remark 4 (The hierarchy).

$$EC \subsetneq EC_\Delta \subsetneq \{\text{all classes of structures}\}.$$

The fundamental limitation: FOL cannot distinguish finite from infinite models. This is not a weakness to be patched — it is the precise boundary of first-order expressibility, and the compactness and Löwenheim-Skolem theorems will make it exact.

The Full Picture

The semantic machinery of FOL operates at three levels.

At the level of *individual structures*, the Tarski satisfaction definition gives a mathematically precise account of when a formula is true — reducing truth to term evaluation followed by set-membership checks, recursing on formula complexity, living entirely in the metalanguage.

At the level of *relations between formulas and structures*, logical implication ($\Gamma \models \varphi$) and validity ($\models \varphi$) measure universal semantic truth — what holds across all structures, not just one.

At the level of *classes of structures*, definability and the EC/ EC_Δ hierarchy measure the expressive power of the language as a whole — which collections of structures can be filtered by sentences, and which lie beyond the reach of FOL.

The completeness theorem — that $\Gamma \models \varphi$ iff $\Gamma \vdash \varphi$ — bridges the semantic side developed here with the syntactic side to be developed in §2.6. It is the central theorem of this chapter.

One Question That Remains Open For Me

The compactness theorem is invoked twice here — to show infinite groups are not EC , and to show finite graphs are not EC_Δ . Both arguments have the same shape: assume a single sentence or finite set works, derive a contradiction using compactness. What is the deep reason compactness has this power? Is it fundamentally a topological fact, an algebraic fact, or something specific to the syntax of FOL?

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